PROJECT REPORT

TASK 21: MISSING VALUE ANALYSIS USING PANDAS

1. Abstract

This project focuses on identifying and analyzing missing values in a numerical and categorical dataset using Python and Pandas. It calculates the number and percentage of missing values in each column, identifies records containing incomplete information, and determines which columns require further treatment.

The project demonstrates the importance of identifying missing data before performing data analysis or developing machine-learning models.

2. Introduction

Missing values are a common problem in real-world datasets. They occur when information is unavailable, not recorded, or incorrectly collected.

Missing data can affect statistical calculations, reduce the quality of analysis, and create problems during machine-learning model training.

Pandas provides useful functions such as `isnull()`, `isna()`, and `sum()` to identify and count missing values.

This project uses a sample student performance dataset to demonstrate missing-value identification and analysis in Google Colab.

3. Objectives

The main objectives of this project are:

- To identify missing values in a dataset.
- To calculate column-wise missing-value counts.
- To calculate missing-value percentages.
- To create a missing-value summary.
- To identify columns that require treatment.
- To examine rows containing missing information.
- To understand the importance of handling missing data before machine learning.

4. Tools and Technologies

Python: Used to develop and execute the analysis.

Pandas: Used to load, inspect, and analyze missing values in the dataset.

NumPy: Used to represent missing numerical observations through NaN values.

Google Colab: Used as the Jupyter Notebook environment.

CSV: Used to store and export the dataset and summary.

5. Dataset Description

The project uses a manually created student performance dataset containing 12 fictional student records.

The dataset includes six columns:

Column	Description
student_id	Unique student identifier
name	Student name
department	Academic department
age	Student age
marks	Marks obtained
attendance	Attendance percentage

Intentional missing values have been introduced in the department, age, marks, and attendance columns to demonstrate the analysis.

The dataset is illustrative and does not contain real student information.

6. Methodology

The project follows these steps:

1. Import Pandas and NumPy.
2. Create a student performance dataset.
3. Introduce missing values using NumPy NaN values.
4. Convert the data into a Pandas DataFrame.
5. Export the DataFrame into a CSV file.
6. Load the CSV file using Pandas.
7. Identify missing values using `isnull()` and `isna()`.
8. Calculate missing-value counts using `sum()`.
9. Calculate missing-value percentages for each column.

10. Create a complete missing-value summary.
11. Identify rows and columns containing missing data.
12. Recommend appropriate treatment methods.

7. Functions Used

7.1 `isnull()`

The `isnull()` function identifies missing values in a DataFrame. It returns True for missing entries and False for available values.

7.2 `isna()`

The `isna()` function is another method for identifying missing values in Pandas. It provides functionality equivalent to `isnull()`.

7.3 `sum()`

The `sum()` function calculates the total number of missing values in each column when applied to a Boolean missing-value DataFrame.

7.4 Missing-value percentage

The missing-value percentage is calculated using:

Missing Percentage = (Missing Values / Total Records) × 100

This indicates the proportion of missing entries in each column.

7.5 Boolean row selection

The expression `df.isnull().any(axis=1)` identifies rows containing at least one missing value.

8. Results and Missing-Value Summary

The analysis identifies the following missing values:

Column	Missing Count	Missing Percentage
student_id	0	0%
name	0	0%
department	1	8.33%
age	1	8.33%
marks	2	16.67%

attendance 2 16.67%

Total rows: 12

Total columns: 6

Total cells: 72

Present values: 66

Missing cells: 6

Overall missing-value percentage: 8.33%.

9. Identification of Columns Requiring Treatment

The analysis identifies four columns containing missing values.

Department

One department entry is missing. Since this is categorical information, possible treatment methods include replacing the missing entry with the most frequent department or creating an Unknown category.

Age

One age value is missing. Median imputation is one possible approach for numerical data.

Marks

Two marks entries are missing. Median imputation can be considered to avoid excessive influence from extreme values.

Attendance

Two attendance entries are missing. Median imputation is another possible approach for this numerical column.

The appropriate treatment method depends on the nature of the dataset and the reason for the missing values.

10. Key Observations

- The dataset contains 12 records and six columns.

- A total of six cells contain missing values.
- The department column has one missing value.
- The age column has one missing value.
- The marks and attendance columns each have two missing values.
- The student_id and name columns have no missing values.
- Six records contain at least one missing value.
- The overall percentage of missing cells is 8.33%.

11. Importance of Missing-Value Analysis

Identifying missing values is an important step in data preparation. Missing information can affect the accuracy of statistical analysis and the reliability of machine-learning models.

By examining missing-value counts and percentages, analysts can decide whether to use imputation, remove incomplete records, or investigate why the data is missing.

12. Applications

Missing-value analysis is useful in:

- Student performance analysis
- Employee and salary datasets.
- Healthcare data analysis.
- Sales and customer datasets.
- Exploratory data analysis.
- Machine-learning data preparation.

13. Limitations

The project uses a small, manually created dataset with intentionally inserted missing values. It focuses on identifying missing information rather than performing complete data cleaning.

The suggested treatment methods are not applied to the original dataset.

14. Future Enhancements

The project could be extended by:

- Applying median and mode imputation.
- Comparing different missing-value treatment methods.
- Visualizing missing values using charts.
- Working with real-world dataset
- Evaluating the impact of missing-value treatment on machine-learning performance.

15. Conclusion

This project demonstrates how Python and Pandas can identify and analyze missing values in a dataset.

It calculates column-wise missing-value counts and percentages, identifies incomplete records, and highlights columns requiring treatment.

The project provides a foundation for understanding data cleaning and preparing reliable datasets for further data analysis and machine learning.

16. References

1. Pandas Documentation: https://pandas.pydata.org/docs/
2. Pandas `isnull()` Documentation: https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.isnull.html
3. NumPy Documentation: https://numpy.org/doc/

18. Project Links

GitHub Repository: https://github.com/venky1318/Missing_Value_Analysis